



Team 2027 Electrical System Technical Specification

Individual and Overall System Requirements and Design Constraints

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1 Purpose

This document describes the requirements that apply to the development of Vortex NTNU's PCBs and overall electrical system. The document shall serve as a common reference for when questions arise regarding what the PCBs shall be capable of, which constraints apply, how they will work with the rest of the system, and which characteristics shall be prioritized.

The requirements in this document primarily describe what the PCBs shall achieve. Details concerning how the requirements are implemented shall be documented in the relevant design documents.

2 Scope

This requirements specification applies to each PCB, describing its:

- Goal
- Mechanical constraints
- Connectivity
- Power
- Signals
- Firmware
- PCB specific requirements

3 Drone Information

This section lists all the already determined drone specs that will be relevant for the PCB specifications and requirements.

3.1 Elhouses

The drone will consist of 3 Bluerobotics elhouses. All three are 300 mm long.

Control-House

One of them will be a 150 mm diameter, 300 in length aluminium housing and will be called the control-house. From top to bottom it will consist of:

- Main connector-plate
- Power-interface PCB
- Sensor-interface PCB
- Two STIM-300 IMUs
- Gigablox ethernet switches (1x Rugged + 1x small switch)
- Orin-interface PCB
- ORIN AGX + ORIN-BOB
- ZED-mini camera
- Clear acrylic plate

The top of the elhouse will have a connector plate, while the bottom consists of a clear acrylic plate.

Actuator-Houses

The other two elhouses will be 75 mm in diameter, 300 mm in length, and made entirely of aluminium with connector-plates on both ends. They will be henceforth called the actuator-houses. From top to bottom they will consist of:

- Main connector-plate
- Torpedo/Marker-dropper PCB (one in each actuator house)
- Battery with BMS PCB
- Motorcontroller PCB
- 4-channel ESC
- Motor connector plate

Everything from the Battery to the ESCs will be mechanically attached to the motor connector plate, such that the battery can easily be removed and replaced. This also means that pogo-pins will be required to facilitate a smooth connection to the rest of the system. This way the batteries along with the motor related electronics can be slotted in to the actuator-house without hassle.

3.2 Batteries

We will be using two 4S-lipo batteries. They are the same ones that where used on Orca. A BMS will be used to ensure they don't charge each other and prevent damage due to overdischarge or overcharge.

3.3 Big Sensors

The drone will be equipped with some quite big and expensive sensors.

- 1x Nortek Nucleus 1000 (DVL)
- 1-2x STIM-300 (IMU) [rumors say we have two]
- 3x FLIR PoE Camera (Camera)
- 1x Sonar (Sonar) [not quite decided]
- 1x Acoustics hydrophone array (Acoustics)

These sensors cost more than your big-stipend so treat them with care. A few of them are also borrowed so we really don't want to break them.

4 Shared Requirements

This section defines the requirements that are shared across all PCBs while noting when exemptions apply.

4.1 Würth Component Use

Requirement

All PCBs wherever possible shall use Würth components in their design. This applies especially to passives like capacitors and resistors.

All Würth design kit components that we will use are available in the "30 – Electronics" Teams channel under "00 – Electrical standards".

Rationale

We receive them for free and already have a design kits for many of them.

4.2 Shared microcontroller

Requirement

All PCBs shall use the STM32C542RCT6 microcontroller.

The following are exempt and have either no microcontroller or an STM32H753VIT6:

- Sensor-interface PCB
- Motorcontroller
- Orin-interface
- Orin-BOB

Rationale

A common microcontroller that costs less than 30kr is nice to use so that when something gets fried we have backups.

4.3 Shared Power Voltage Range

Requirement

All PCBs shall be capable of taking in the battery voltage directly.

Rationale

This ensures we only have one power net and prevents the need for a PDB.

4.4 Shared 3.3V regulator

Requirement

All PCBs shall use the same Würth 173950336 3.3V switching mode regulator unless space constraints or special permissions apply.

Rationale

This ensures that we have replacements for when they inevitably get fried the day before the competition.

4.5 Shared eFuse and power protection

Requirement

All PCBs and their components shall have their power inputs protected by one or multiple eFuses. It is required for all systems (like microcontrollers) pulling less than 2 Amps to use the TPS26400PWPR eFuse.

Rationale

We have had nasty experiences with stupid mistakes costing us entire PCBs. This mainly includes overvoltage and reverse polarity problems due to user error during testing. The TPS26400PWPR comes with all the features we would want and more to protect our circuits from future mistakes.

4.6 Shared UVLO and OVLO eFuse settings

Requirement

The TPS26400PWPR is configured in a hardwired fashion with resistors and capacitors. The same values and components shall be used to ensure that UVLO and OVLO is the same for all PCBs. Notice that overcurrent and softstart will need to be configured on an individual PCB basis.

Rationale

This ensures that no systems are powered during an overvoltage or undervoltage fault on the main power lines. This is especially important for undervoltage such that we don't overdischarge the batteries.

4.7 Current monitoring

Requirement

All PCBs must be capable of measuring the total current draw of themselves and any connected device which cannot measure its own current draw. This can be done through the eFuse or dedicated current sensors. It is not necessary to have individual current sensors for each connected device/system, so long as the total current draw for the PCB and its connected devices is known.

Rationale

It is a part of the specification to know the effect draw of the drone. Being able to know the current draw of various systems is also useful in determining faults when ICs are fried.

4.8 Shared CAN-transceiver and CAN-bus circuitry

Requirement

All PCBs that interface with the CAN-bus must use the same TCAN3414DR or ADM3057 can-transceiver. The TCAN3414DR is simpler and easier to handle but is not isolated. The ADM3057 is isolated and must therefore be used on the following:

- Power-interface (specifically the actuator CAN-bus connection)
- BMS
- Orin-BOB

Rationale

We have had many experiences with frying CAN-transceivers so using the same one will allow us to easily replace them. Isolated CAN-transceivers should also allow us to avoid frying them in the first place.

4.9 CAN peripherals and pins on the STM32C542

Requirement

All STM32C542s must use the FDCAN2 peripheral and have it use PB5 and PB6 pins for CANRX/TX.

Rationale

This allows us to use the built in bootloader.

4.10 USB interface

Requirement

All PCBs with microcontrollers need to be equipped with a USB C connector such that the PCBs can be powered and interfaced with via a USB cable. The interface must be either done directly (STM32s tend to support this) or via a USB-to-UART chip. It does NOT need to support the fastest USB speeds.

Rationale

A USB connector gives us a way to power and test the PCBs without needing 6-7 power-supplies. This is nice because we do not have 6-7 power supplies.

4.11 UART interface

Requirement

All PCBs with microcontrollers need to be equipped with an exposed 2 pin UART interface.

Rationale

If CAN-bus fails then we have a backup. And it can be used for debugging.

4.12 LED-indicators

Requirement

All PCBs need to be equipped with a set of diffused red, yellow, green and blue LEDs. The blue ones are directly powered from the 3.3V, while the rest are controlled by the microcontroller. There must also be a red LED controlled by the eFuse fault/PG indicator.

Rationale

This gives us a way to easily see the PCBs status and power state.

4.13 Programming/debugging

Requirement

All PCBs with microcontrollers must come with a full SWD interface for programming and debugging.

Rationale

We need this in order to program and debug.

4.14 Reset/Boot buttons

Requirement

All PCBs with microcontrollers must come with reset and boot buttons.

Rationale

This is so we can reset and boot the microcontrollers.

4.15 HSE Oscillator

Requirement

All PCBs with microcontrollers must be equipped with a 830207659609 and have it connected as a digital bypass for the internal clocks.

Rationale

It's a High speed external oscillator for the microcontroller. It generates a square wave at 48 MHz.

4.16 Wiring Color Convention

Requirement

Internal and external wiring shall follow a consistent vehicle-wide color convention.

At minimum:

- Red: Positive battery voltage
- Orange: Voltages other than V-batt
- Black: Ground / power return
- Blue / White: CAN-H / CAN-L
- Brown / White: RS485 or RS422 differential pair
- Green / Yellow: UART TX / RX
- Purple: PWM or control signals

Note that ethernet is exempt from this color-coding.

Reserved colors shall not be used for unrelated functions.

Rationale

A consistent color convention simplifies assembly, inspection, and troubleshooting and reduces the risk of incorrect electrical connections.

4.17 ESD protection

Requirement

All connectors and exposed test points must have TVS diodes on the connector inputs.

Rationale

To avoid fry by touch.

5 Control-house PCBs

This section defines the requirements for PCBs that will be in the control-house.

5.1 PCB Tower

Requirement

All PCBs in the control-house will be arranged in a PCB tower such that they are stacked like Oreos™. It must however be possible to disconnect individual PCBs without needing to disassemble the entire thing.

Rationale

This is due to space constraints. Being able to disconnect each PCB individually is necessary to facilitate easy work on the PCBs.

5.2 Shared PCB Outline

Requirement

All PCBs in the control-house must have a shared common outline. It must be MAX 140 mm in diameter and must facilitate the possibility to slide out the PCBs in the middle without undoing the entire tower. The outline must also facilitate the common connectors that are shared among the control-house PCBs.

Rationale

This is to ensure that mechanical won't need to facilitate each design separately and can instead facilitate one outline.

5.3 Shared Inter Board Connector

Requirement

All PCBs in the control-house must have a shared common inter-board connector. This will be a 691378140024 on the PCB with a 691379180024 to plug in. Power and CAN-bus will run on the same pins and be daisy chained from each PCB to the next. Any other signals that go to other components in the elhouse should go through this one.

Rationale

These connectors are pluggable "screw terminals". The fact that they are screw terminals gives us some freedom in how to connect them and avoids the need for annoying crimps like we have used before (though cables must still be crimped with ferrules but that's easy). Since they are pluggable we simply need to plug the entire thing out if we want to remove the PCB from our tower. This should hopefully avoid the need to disconnect 9000 different connections and then remember how to connect it back each time we want to remove a PCB.

6 Shared Control-house PCB Requirement

This section defines the requirements for the shared PCB outline that will be in the control-house.

6.1 Max diameter

Requirement

The PCB shall fit entirely within a circle with a diameter of 135 mm

Rationale

We need space for 3D prints along the edges and it needs to fit through the flange.

6.2 Cable channel

Requirement

There needs to be a cable channel that is 42 mm wide and 40 mm deep at the top of the PCB

Rationale

We need space both for the shared screw terminal and the cables passing from layer to layer.

6.3 Cut side edges

Requirement

1 cm shall be cut off from the two sides of the PCB circle such that they are straight.

Rationale

On the left and right side of the PCB there needs to be a thick print that actually carries the weight from the upper layers. The space occupied by the cut will be dedicated to this load bearing print.

6.4 No place zone

Requirement

From the straight cut side edges of the PCB there shall be a 5mm wide no place zone

Rationale

For the load bearing print to be able to hold the PCBs there needs to be space for a shelf type print that goes over and under the PCB on each side. This way we can simply slide the PCBs into it like shelves on in a cupboard.

7 Actuator-house PCBs

This section defines the requirements for PCBs that will be in the actuator-houses.

7.1 Board dimensions

Requirement

All PCBs in the actuator-house must fit in the actuator house.

Rationale

I leave the rationale of this requirement as an exercise to the reader.

8 CAN-bus

This section defines the shared requirements for the various CAN-bus interfaces. The CAN-IDs that these will run on shall be decided when we allocate CAN-IDs.

8.1 Startup/checkup message

Requirement

All systems must send a startup message. If a message with this same ID is received from a different system, a response consisting of the startup message on that ID must be sent.

Rationale

This is to ensure that we can check which systems are working.

8.2 Restart command

Requirement

All systems must have a CAN-ID dedicated as a restart command, such that when a message is sent on their restart ID the corresponding system will restart. This functionality must be implemented so such that it will work even if the firmware is stuck in a loop.

Rationale

This is to ensure that we can restart systems that aren't responding.

8.3 Example of System Checkup

Here are two examples of exchanges over the CAN-bus during startup and later during a checkup.

- "BMS OK" sent at ID 0x100 from the BMS at startup.
Some time passes...
- "Checkup" sent from Power-interface at ID 0x100 due to no BMS activity on the CAN-bus.
- "BMS OK" sent at ID 0x100 from the BMS due to checkup.

An alternative interaction where no response arrives and the power-interface requests a restart.

- "BMS OK" sent at ID 0x100 from the BMS at startup.
Some time passes...
- "Checkup" sent from Power-interface at ID 0x100 due to no BMS activity on the CAN-bus.
No reply...
- "restart" sent at ID 0x101 from the Power-interface due to failed checkup.

- "BMS OK" sent at ID 0x100 from the BMS due to starting up again.

9 ORIN

This section defines the shared requirements for any systems that interface with or power the ORIN.

9.1 No external reset/power shutoff

Requirement

With the exeption of the BMS, no system should be capable of shutting off or resetting the ORIN during non fault states.

Rationale

It would be very problematic if the ORIN shuts off or resets randomly during an autonomous run.

9.2 Protection

Requirement

MAKE SURE THERE IS NO WAY THIS THING GETS FRIED.

Rationale

Its coming out of your big stipend.

10 Killswitch

This section defines the shared killswitch requirements for any systems that provide power to mechanical actuators.

10.1 Actuator killing

Requirement

When the killswitch is activated all power to actuators must be cut. This shall be done in a hardcoded, hardwired fashion.

Rationale

Robosub requires a way to shut-off all mechanical actuators. Doing this in a hardwired fashion ensures that no software bugs can interfere with this requirement.

10.2 Inter board signal active state vs normal state

Requirement

When the killswitch is activated the killswitch signal shall be floating. When not active it shall be connected to GND.

Rationale

GND and V-batt are the most robust shared reference points for any system on the drone. By using them instead of say a 3.3V signal, we avoid having to worry about the voltage difference between the different 3.3V supplies on the different PCBs.

10.3 Optocoupler interface

Requirement

All systems that use the killswitch signal must do so through an optocoupler. How the killswitch signal is utilized on the other side of the optocoupler is board specific.

Rationale

This ensures that the killswitch signal is current controlled, such that noise on the signal line has no way to activate the actuator shut-off sequence. This also allows for the receiving board to adapt the signal such that it shuts off the actuators in whatever way is convenient for that actuator.

11 Power-interface PCB

The following is a more in-depth description of the objective and requirements of the power-interface PCB.

11.1 Objective

The power-interface PCB will act as the main interface between the connector plate and the rest of the control-house. It's main job is to facilitate connections to various outside systems like the actuator houses, lights, gripper and so on. It will also be responsible for monitoring these systems in different ways like their current draw and CAN-bus activity. The power-interface PCB will also be responsible for the killswitch.

11.2 Connections

Requirement

The PCB must support at the very least the following connections:

The ones marked [IDK] are unknown and up to the designer to decide on/figure out. The 16 free pins on the screw terminal are available for use to connect to anything that is on the control-house.

Note that power and CAN only use 8 of the 24 pins on the screw-terminal. The 8 pins remaining for misc available for board specific use.

Rationale

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
Screw-terminal	Control-house	24	Power, CAN (16 free)	Interface with control-house
[IDK]	Camera lights	3	Power and PWM	So we can control the lights
[IDK]	Indicator lights	4	Power and Data	So we can control the lights
[IDK]	Killswitch	2	Killswitch	Interface killswitch
[IDK]	Actuator-house 1	6	Power, CAN, killswitch	Interface actuator house 1
[IDK]	Actuator-house 2	6	Power, CAN, killswitch	Interface actuator house 2
[IDK]	Gripper-PCB	6	Power, CAN, killswitch	Interface gripper

These are all planned to be a part of the drone and therefore need corresponding connectors.

11.3 Current monitoring

Requirement

It must be capable of individually monitoring the current draw of itself, the control-house and the current draw from each of the actuator houses.

Rationale

Being able to know the current draw of the control-house is a nice to know. Knowing its own current draw is a part of the shared requirements, and knowing the current draw from each of the actuator houses is important to know in order to determine if one of the actuator houses is being drained more than the other.

11.4 CAN-bus monitoring

Requirement

It must be capable of monitoring the other PCBs activity on the CAN-bus. If another PCB has remained silent for a long time it will be the power-interfaces job to do a checkup over the can-bus. If no reply is received during the check-up it must report this to the GUI and attempt to reset that system. The reset should be done over CAN-bus (even if their CAN-bus doesn't work).

Rationale

Being able to restart systems over the CAN-bus is necessary to facilitate a checkup.

11.5 CAN-bus logging

Requirement

All CAN-messages must be logged to an easily accessible non volatile storage medium like an SD-card.

Rationale

This will allow us to review what happened on the CAN-bus during a tether-less run. Like during Robosub.

11.6 Power protection

Requirement

Actuator-house power inputs must have reverse-polarity protection. All power outputs must have short circuit protection.

Rationale

Reverse polarity protection ensures that we don't fry ourselves. Short circuit protection on the outputs ensures that shorting the circuit doesn't fry the power-interface.

11.7 Gripper support

Requirement

The power-interface must provide power, CAN-bus and a killswitch signal to the gripper.

Rationale

The gripper needs it.

11.8 Camera light support

Requirement

The power-interface must provide power, and a microcontroller controlled PWM signal to the camera lights. The PWM signal should be set based on messages from the CAN-bus.

Rationale

We need a way to power and control the camera lights.

11.9 Indicator light support

Requirement

The power-interface must provide power and control signals to the indicator light such that they can be controlled. The indicator light should reflect the state of the drone and how this is achieved is left to the members.

Rationale

We need a way to power and control the indicator lights so that we can determine the drone state by looking at it.

11.10 Killswitch support

Requirement

The killswitch must be connected such that the killswitch specifications are met.

Rationale

The specs must be met.

11.11 Telemetry

Requirement

This PCB must be equipped with a BME280 pressure, temperature and humidity sensor. The measurements must be posted on to the CAN-bus at atleast 1 Hz

Rationale

It is important to be able to know the temperature and pressure inside the el-houses.

11.12 Ideal Diode

Requirement

The actuator house power inputs must have ideal diodes.

Rationale

To prevent the batteries from charging each other.

12 Sensor-interface PCB

The following is a more in-depth description of the objective and requirements of the sensor-interface PCB.

12.1 Objective

The sensor-interface PCB will act as the main interface between our motion sensors and and the rest of the drone. It will power, configure, monitor and read the sensors data and run a sensor fusion algorithm to provide an estimate of the drones position, velocity, acceleration and orientation.

The main sensors it will interface with are:

- Norteks Nucleus 1000 DVL
- STIM-300 IMU
- Bluerobotics Bar02 Pressure sensor
- Magnetometer

Nucleus 1000

The nucleus has a built in IMU, magnetometer, pressure sensor and most importantly a DVL. It performs it's own sensorfusion with these and gives us its own estimate of our drones position, orientation and so on. We ultimatly want to fuse the nucleus's fused output with our own sensors. This way we utilize the nucleus's built in sensors in our final estimate. Note that i the writer have not taken sensor fusion so this might be completely wrong, do your own research or talk to someone who has taken the subject or is currently taking it for better information regarding this matter.

In addition to doing its own sensorfusion, the nucleus allows us to read its sensor data raw. The one we care about is the DVL. It measures the drones speed relative to the surface using the doppler effect.

STIM 300

The STIM 300 is our IMU. It is significantly better than the Nucleus's built in IMU, which is precisely why we want to use it. It measures our xyz-acceleration and rotational velocity along the xyz-axes.

Bar02

The Bar02 is a pressure sensor. It measures the outside water-pressure. This is quite useful since the pressure is directly proportional to our depth.

Magnetometer

Excluding the nucleus's built-in magnetometer, we currently don't have a magnetometer on the drone. As such you must find one and integrate it into the system. The magnetometer is used literally as a compass. The IMU can tell us which way down is due to gravity but it has no way of keeping perfect track of which way we are facing in the xy-plane.

12.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
Screw-terminal	Control-house	24	Power, CAN (16 free)	Interface with control-house
[IDK]	STIM-300	12	Power, RS422, Trig, misc	Interface with STIM-300
[IDK]	Nucleus	8	Power, RS422, Trig, misc	Interface with Nucleus 1000
[IDK]	Bar02	4	Power, I ² C	Interface with pressure sensor

The ones marked [IDK] are unknown and up to the designer to decide on/figure out. The 16 free pins on the screw terminal are available for use to connect to anything that is on the control-house.

Note that power and CAN only use 8 of the 24 pins on the screw-terminal. The 8 pins remaining for misc available for board specific use.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connectors.

12.3 Sensor power-supplies

Requirement

The PCB must be capable of powering the sensors it interfaces with.

Rationale

They don't work without power.

12.4 Power-protection

Requirement

The sensor power lines must be well protected. Short circuit protection and overvoltage protection are musts.

Rationale

They are really expensive and must therefore be protected.

12.5 Interface-protection

Requirement

The sensor digital signals must be galvanically isolated from the microcontroller. Consider using Würth's digital isolators.

Rationale

If the 3.3 V net or microcontroller is fried then the sensors will remain unaffected.

12.6 Handling of "optional" sensor signals

Requirement

All sensor signals that can be used to achieve a better sensor fusion result or a better product in general **MUST** be used. This includes but is not limited to interrupt pins, trigger pins and configuration pins.

Rationale

They are there for a reason. Don't omit them just because they aren't needed for the bare minimum functionality.

12.7 Sensor data handling

Requirement

The sensor data should be fused on the microcontroller. If this turns out to be an impossibility, the data must at the very least be passed on to the ORIN.

Rationale

We need a working ESKF please.

13 ORIN-interface PCB

The following is a more in-depth description of the objective and requirements of the ORIN-interface PCB.

13.1 Objective

The ORIN-interface PCB will act as the main interface between the ORIN and the rest of the drone. It's main job is to facilitate connections like CAN-bus and power to the ORIN. It will also be hosting a GigaBlox Nano 1 gigabit ethernet switch as well a a 4 channel PoE injector.

13.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
Screw-terminal	Control-house	24	Power, CAN (20 free)	Interface with control-house
DC-Jack	ORIN	2?	Power	ORIN Power
[IDK]	ORIN-BOB	40	Miscellaneous	ORIN Pinout
[IDK]	Rugged Gigablox	4	Power	Ethernet switch power
[IDK]	Small Gigablox	4	Power	Ethernet switch power
[IDK]	Rugged Gigablox	8	Ethernet "in"	FLIR Camera data
[IDK]	Rugged Gigablox	8	Ethernet "in"	FLIR Camera data
[IDK]	Small Gigablox	8	Ethernet "in"	FLIR Camera data
[IDK]	Small Gigablox	8	Ethernet "in"	Acoustics data
[IDK]	Acoustics	8	Ethernet, PoE out	Acoustics power and data
[IDK]	FLIR Camera	8	Ethernet, PoE out	Camera power and data
[IDK]	FLIR Camera	8	Ethernet, PoE out	Camera power and data
[IDK]	FLIR Camera	8	Ethernet, PoE out	Camera power and data

The ones marked [IDK] are unknown and up to the designer to decide on/figure out. The 20 free pins on the screw terminal are available for use to connect to anything that is on the control-house.

Note that power and CAN only use 4 of the 24 pins on the screw-terminal. The 20 pins remaining for misc available for board specific use.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connections.

13.3 Orin pinout exposure

Requirement

A UART, SPI, I2C and CAN interface must be exposed via the standard screw terminal. Same applies to 5V, 3.3V and GND.

Rationale

In case a quick fix for something is needed.

13.4 Orin Power

Requirement

The Orin is powered via a DC-Jack. It should accept the battery voltage, however it must go through an eFuse or similar first.

Rationale

It needs power and we don't want to fry it.

13.5 Gigablox Power

Requirement

The Gigablox is powered via a connector. It should accept the battery voltage, however it must go through an eFuse or similar first.

Rationale

It needs power and we don't want to fry it.

13.6 PoE Ports

Requirement

The ORIN-interface must have four 1 gig ethernet ports supporting PoE. For each PoE there must be a corresponding input and output.

Rationale

We have four PoE devices.

13.7 PoE negotiation

Requirement

The PoE must support 1 Gb/s speeds and do some sort of negotiation. Aka it must support 4PPoE

Rationale

There was once a FLIR camera that got fried by a dumb PoE injector. We would like to avoid this in the future

13.8 Info regarding Cameras

The FLIR cameras run 1440x1080 at 20 fps. Each pixel gives an 8 bit color value raw which results in a 250 Mb/s data rate PER CAMERA. If we want RGB and not just the raw data this turns into 750 Mb/s!

13.9 Mounting

Requirement

This PCB will be mounted to the ORIN directly, using the existing screw holes.

Rationale

It will not be a standard PCB layout.

14 ORIN-BOB PCB

The following is a more in-depth description of the objective and requirements of the ORIN-BOB (Break Out Board) PCB.

14.1 Objective

The ORIN-BOB is responsible for breaking out the ORIN 40-pin pin header in a safe and isolated manner.

14.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
Pin-header	ORIN-devkit	40	Power and misc	The source for the signals that will be broken out
[IDK]	ORIN-interface PCB	[IDK]	Communication protocols	Forward communication protocols

The ones marked [IDK] are unknown and up to the designer to decide on/figure out.

Rationale

These are all planned to be a part of the PCB and therefore need corresponding connectors.

14.3 CAN-bus

Requirement

This PCB must be equipped with a standard CAN-transceiver.

Rationale

The most important part of this board is breaking out the CAN-bus for the ORIN. Orin doesn't have a CAN-transceiver. Only a can controller.

14.4 Protocols to break out

Requirement

The following protocols must have a their signals available:

- 2x CAN (CAN-L, CAN-H)
- UART (TX, RX)
- SPI (SCLK, MOSI, MISO, CS)
- I²C (SDA, SCL)

Rationale

In case we need to last minute bodge a sensor into the drone, it suddenly becomes nice to have protocols available.

14.5 Digital Isolation

Requirement

All pins on the ORIN must be galvanically isolated from the output pins.

Rationale

We don't want to fry ORIN

15 Torpedo/Marker-dropper controller PCB

The following is a more in-depth description of the objective and requirements of the Torpedo/Marker PCB.

15.1 Objective

The Torpedo/Marker PCB is responsible for monitoring, powering and controlling the actuators that control the torpedo and marker-dropper system.

15.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
[IDK]	Power-interface	6	Power, CAN, killswitch	Interface control-house
Pogo-pads	BMS	6	Power, CAN, killswitch	Interface BMS
[IDK]	Bluetrail Servo	3-4	Power and CAN/PWM	Servo connection

The ones marked [IDK] are unknown and up to the designer to decide on/figure out.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connectors.

15.3 Servo power input protection

Requirement

Input power must have reverse polarity and UVLO functionality for the servos power input

Rationale

To prevent the servo from dying.

15.4 Servo power output protection

Requirement

Current overload protection and short circuit protection is required for the power lines going to the servo.

Rationale

To prevent the servo from dying.

15.5 Current monitoring

Requirement

Must be capable of measuring the servos current draw.

Rationale

It will be used to estimate the servos outputted torque.

15.6 Telemetry

Requirement

This PCB must be equipped with a BME280 pressure, temperature and humidity sensor. The measurements must be posted on to the CAN-bus at atleast 1 Hz

Rationale

It is important to be able to know the temperature and pressure inside the el-houses.

15.7 IMU

Requirement

The PCB must be equipped with a WSEN-ISDS 6 Axis IMU.

Rationale

This is so we can compensate for the fact that the main STIM-300 IMU that is in the control-house cannot be placed at the center of mass.

15.8 Killswitch

Requirement

Active killswitch must kill all power to the servos.

Rationale

It is required by the killswitch spec.

16 BMS PCB

The following is a more in-depth description of the objective and requirements of the BMS PCB.

16.1 Objective

The BMS PCB is responsible for monitoring, managing and protecting the batteries. This includes preventing overdischarge, short circuit and any other thing that would damage the batteries.

16.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
[IDK]	Battery	2	Power	Provide the main battery connection
[IDK]	Battery	5?	Analog voltage	Cell monitoring
Pogo-pins	Torpedo/Marker-dropper	6	Power, CAN, killswitch	Interface Torpedo/marker
[IDK]	Motorcontroller	2	Killswitch	Killswitch connection
[IDK]	Motorcontroller	2	CAN-bus	CAN-bus connection
[IDK]	Motorcontroller	2	Power	Power connection

The ones marked [IDK] are unknown and up to the designer to decide on/figure out.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connectors.

16.3 Current monitoring

Requirement

Must be capable of measuring the battery current draw.

Rationale

It will be used to estimate the remaining battery time.

16.4 Battery voltage measurement

Requirement

Must be capable of measuring the battery voltage.

Rationale

It will be used to estimate the battery charge.

16.5 BMS IC

Requirement

The same IC as was used last year will be used.

Rationale

We have good documentation and last years design can be used as reference.

16.6 BMS IC SPI resistors

Requirement

The SPI lines must have series resistors

Rationale

To avoid sharp rise times and thereby noise.

16.7 Motorcontroller power connector

Requirement

Make sure to use a keyed connector for the power.

Rationale

That way it isn't possible to connect the power in reverse.

16.8 Fault warning

Requirement

If a fault is detected by the BMS IC, the microcontroller must notify the rest of the system of this.

Rationale

That way we know what is going on.

16.9 Power efficiency

Requirement

The microcontroller and PCB must minimize power consumption to the furthest reasonable degree.

Rationale

The BMS will always have power, even when the battery is empty. As such the PCB should consume as little power as possible.

17 Motorcontroller PCB

The following is a more in-depth description of the objective and requirements of the Motorcontroller PCB.

17.1 Objective

The Motorcontroller PCB is responsible for monitoring, powering and controlling the drones thrusters. The motorcontroller must calculate the most optimal way to allocate thrust such that desired force and torque is achieved on the drone. It achieves this by giving power and control commands to a 4-channel ESC running AM32 firmware.

17.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
[IDK]	BMS	2	Killswitch	Killswitch connection
[IDK]	BMS	2	CAN-bus	CAN-bus connection
[IDK]	BMS	2	Power	Power connection
[IDK]	ESC	2	Power	Power connection
[IDK]	ESC	4	PWM/Dshot	Thruster control
[IDK]	ESC	2	[IDK]	Telemetry

The ones marked [IDK] are unknown and up to the designer to decide on/figure out.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connectors.

17.3 Current monitoring

Requirement

It must be possible to monitor the total and individual current draw of the thrusters.

Rationale

They draw the most current so knowing how much they draw is important.

17.4 Voltage monitoring

Requirement

It must be possible to monitor the battery voltage with the microcontrollers built in ADC.

Rationale

The motors thrust is

17.5 Optimal thruster allocation

Requirement

The motorcontroller will receive a desired thrust and torque vector from the CAN-bus. It must then optimally allocate thrust to the thrusters such that this is achieved.

Rationale

It's more optimal, it's literally in the name.

17.6 Killswitch

Requirement

Active killswitch must kill all power to the ESCs and thrusters.

Rationale

It is required by the killswitch spec.

18 Gripper PCB

The following is a more in-depth description of the objective and requirements of the Gripper PCB.

18.1 Objective

The Gripper PCB is responsible for monitoring, powering and controlling the drones gripper arm.

18.2 Connections

Requirement

The PCB must support at the very least the following connections:

Connector type	Connected to	Nr of pins	Type	Connector purpose
USB C	Nothing	24	USB 2.0	Testing and debugging
Pin-Header	Nothing	2	UART	Debugging and backup
SWD-Header	Nothing	10	SWD	Programming and debugging
[IDK]	Power-interface	6	Power, CAN, killswitch	Interface control-house
[IDK]	Bluetrail Servo 1	3-4	Power and CAN/PWM	Servo connection
[IDK]	Bluetrail Servo 2	3-4	Power and CAN/PWM	Servo connection
[IDK]	Bluetrail Servo 3	3-4	Power and CAN/PWM	Servo connection

The ones marked [IDK] are unknown and up to the designer to decide on/figure out.

Rationale

These are all planned to be a part of the drone and therefore need corresponding connectors.

18.3 Current monitoring

Requirement

It must be possible to monitor individual current draw of the servos.

Rationale

It is necessary to determine if they have stalled.

18.4 Optimal thruster allocation

Requirement

Inverse kinematics, it can do it.

Rationale

To control it you know

18.5 Killswitch

Requirement

Active killswitch must kill all power to gripper arm.

Rationale

It is required by the killswitch spec.

19 Deviations and Changes

The requirements in this document may be changed if the project's objectives, competition rules, or technical assumptions change.

Changes shall:

- Be discussed with the relevant technical leads
- Have a clear rationale
- Be recorded in the document's change log
- Be communicated to the affected subsystems